

Switch spoofing attack (vlan hopping attack)

It is a technical method used by attackers to "hop" between different Virtual Local Area Networks (VLANs) without restrictions or authorization. This allows them to access and fully monitor data traffic across the network.

How does this attack work?

In this documentation, I will explain the attack through Switch Spoofing. In this method, we trick the primary switch into believing that our device is another switch rather than a standard computer. Consequently, we obtain a Trunk Link instead of an Access Link. This process is executed by sending DTP (Dynamic Trunking Protocol) packets from the attacker's device to the primary switch. Upon receiving these packets, the switch assumes the attacker's device is a legitimate switch and converts the port mode to Trunk.

Why is this attack dangerous?

This attack is critical because it enables the attacker to monitor and eavesdrop on sensitive data in other departments, such as Finance or Human Resources. Furthermore, it is notoriously difficult to detect.

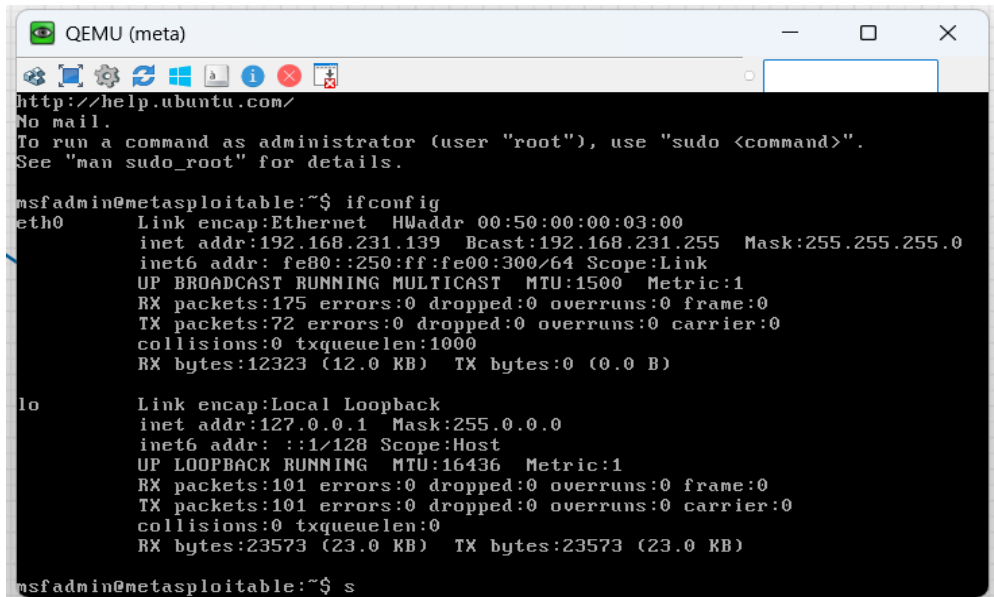
How to prevent it?

Prevention against these types of attacks must be proactive. During the network configuration phase, all ports should be manually configured, and default switch settings such as Auto-Trunking must be disabled.

Will show the attack in our environment step by step :

Firstly: for this attack I will start three devices (kali,metasploitable,switch)

The next figure will show us metasploitable configuration:



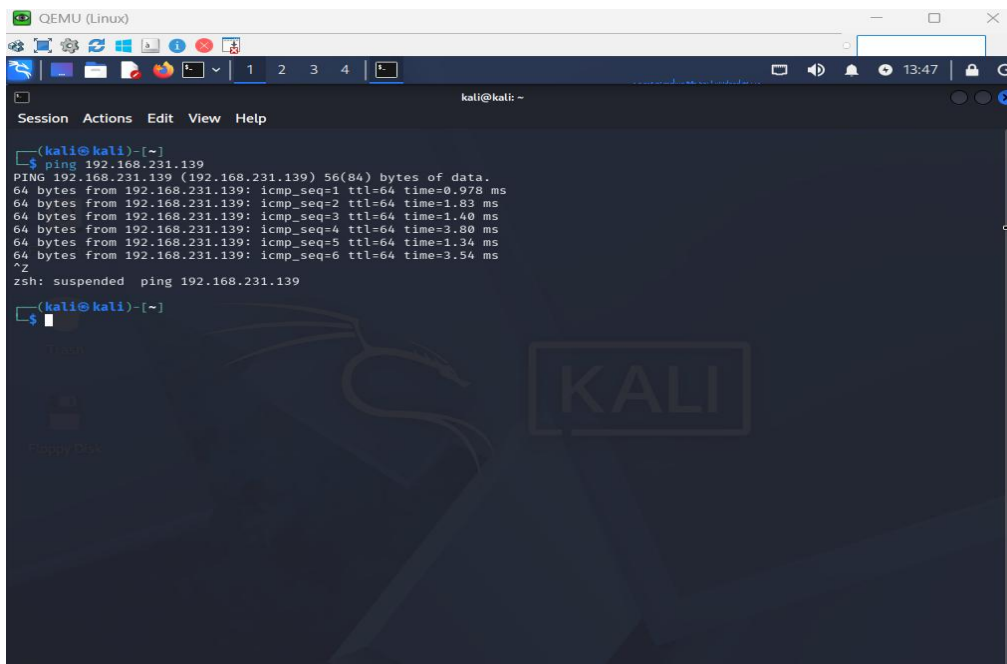
```
QEMU (meta)
http://help.ubuntu.com/
No mail.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:00:00:03:00
          inet addr:192.168.231.139  Bcast:192.168.231.255  Mask:255.255.255.0
          inet6 addr: fe80::250:ff:fe00:300/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:175 errors:0 dropped:0 overruns:0 frame:0
          TX packets:72 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:12323 (12.0 KB)  TX bytes:0 (0.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:101 errors:0 dropped:0 overruns:0 frame:0
          TX packets:101 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:23573 (23.0 KB)  TX bytes:23573 (23.0 KB)

msfadmin@metasploitable:~$ s
```

I took metasploitable ip and I did ping from kali:



```
QEMU (Linux)
kali@kali: ~
Session Actions Edit View Help

(kali@kali)~$ ping 192.168.231.139
PING 192.168.231.139 (192.168.231.139) 56(84) bytes of data:
64 bytes from 192.168.231.139: icmp_seq=1 ttl=64 time=0.978 ms
64 bytes from 192.168.231.139: icmp_seq=2 ttl=64 time=1.83 ms
64 bytes from 192.168.231.139: icmp_seq=3 ttl=64 time=1.40 ms
64 bytes from 192.168.231.139: icmp_seq=4 ttl=64 time=3.80 ms
64 bytes from 192.168.231.139: icmp_seq=5 ttl=64 time=1.34 ms
64 bytes from 192.168.231.139: icmp_seq=6 ttl=64 time=3.54 ms
^Z
zsh: suspended ping 192.168.231.139

(kali@kali)~$
```

As we see the ping was done seccesfully,becaouse both of them in the same vlan as we will see in the next figure.

```
switch1
1    default                active    Et0/0, Et0/1, Et0/2, Et0/3
10   VLAN0010              active
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default       act/unsup
1005 trnet-default         act/unsup
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 12
Switch(config-vlan)#^Z
Switch#
*Apr 21 20:54:38.926: %SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief

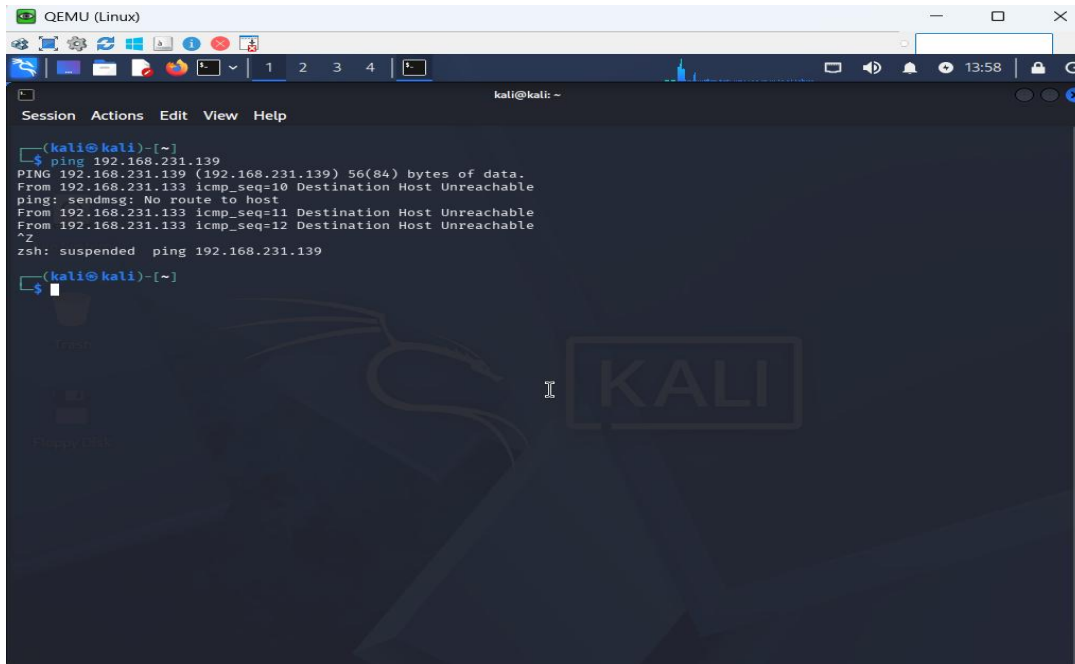
VLAN Name                Status    Ports
-----
1    default                active    Et0/0, Et0/1, Et0/2, Et0/3
10   VLAN0010              active
12   VLAN0012              active
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default       act/unsup
1005 trnet-default         act/unsup
Switch#
```

Now I will create new vlan (vlan 12) and iwill move kali to it.

```
switch1
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default       act/unsup
1005 trnet-default         act/unsup
Switch#
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface e0/1
Switch(config-if)#switchport access vlan 12
Switch(config-if)#^Z
Switch#
*Apr 21 20:57:28.897: %SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Et0/0, Et0/2, Et0/3
10   VLAN0010              active
12   VLAN0012              active    Et0/1
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default       act/unsup
1005 trnet-default         act/unsup
Switch#
```

As we can see now kali in another vlan , lets try ping metaspolaitable.

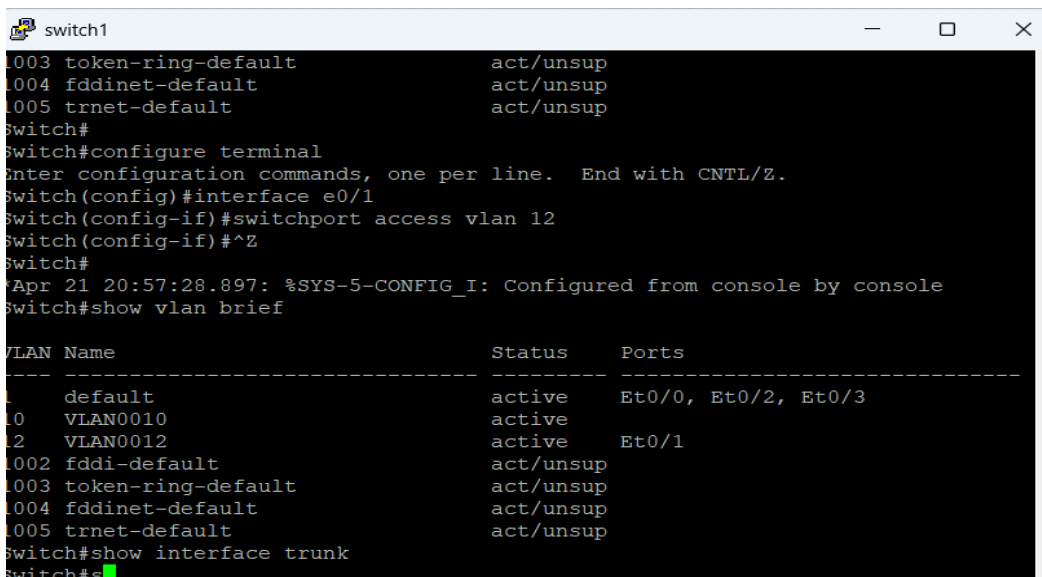


```
QEMU (Linux)
kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ ping 192.168.231.139
PING 192.168.231.139 (192.168.231.139) 56(84) bytes of data.
From 192.168.231.133 icmp_seq=10 Destination Host Unreachable
ping: sendmsg: No route to host
From 192.168.231.133 icmp_seq=11 Destination Host Unreachable
From 192.168.231.133 icmp_seq=12 Destination Host Unreachable
^Z
zsh: suspended ping 192.168.231.139
(kali@kali)-[~]
```

Dosent work right?? That's becaouse kali cant see metaspolaitable sense kali is in another vlan .

And now our task it makes kali able to see all other vlans , and this is the hart of our attak , iwill excute this attack using (yersinia) it's a tool that will help me send DTP packet to the switch .

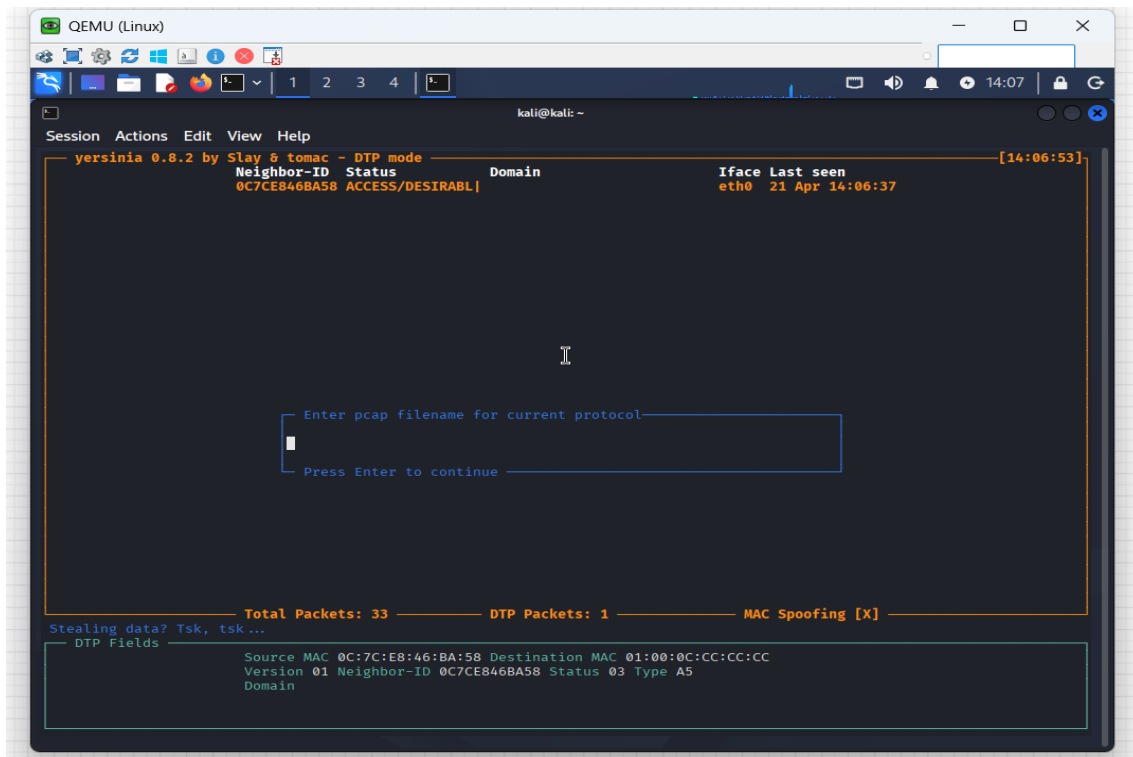
Now I will show you that we don't have any trunk port in the main switch :



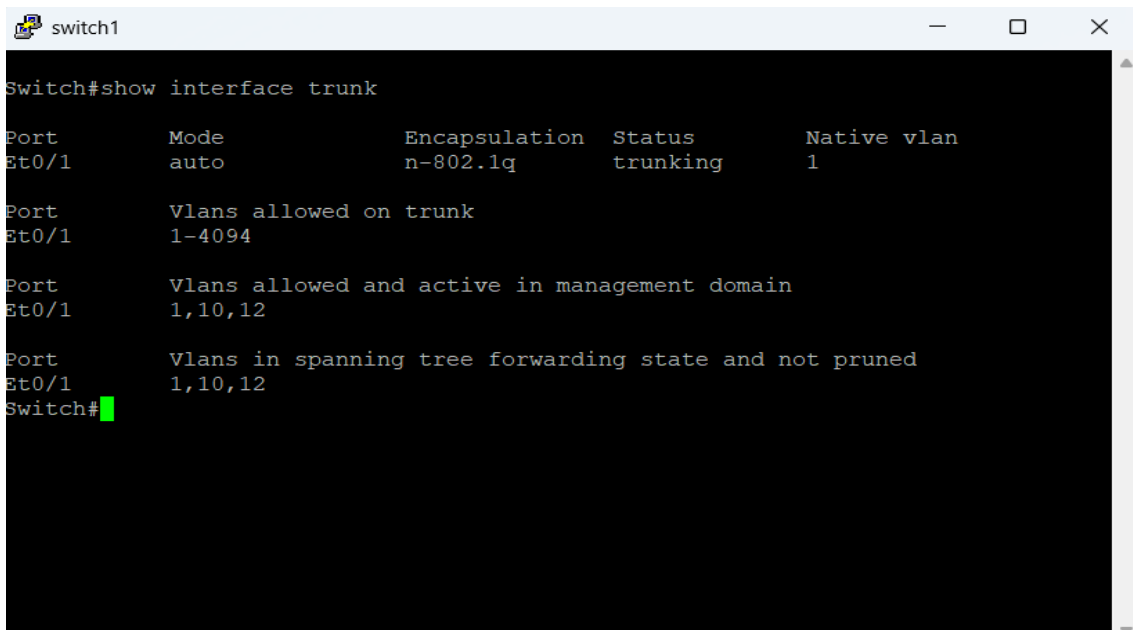
```
switch1
1003 token-ring-default          act/unsup
1004 fddinet-default             act/unsup
1005 trnet-default               act/unsup
Switch#
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface e0/1
Switch(config-if)#switchport access vlan 12
Switch(config-if)#^Z
Switch#
Apr 21 20:57:28.897: %SYS-5-CONFIG_I: Configured from console by console
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default              active    Et0/0, Et0/2, Et0/3
10   VLAN0010             active
12   VLAN0012             active    Et0/1
1002 fddi-default         act/unsup
1003 token-ring-default   act/unsup
1004 fddinet-default      act/unsup
1005 trnet-default        act/unsup
Switch#show interface trunk
Switch#
```

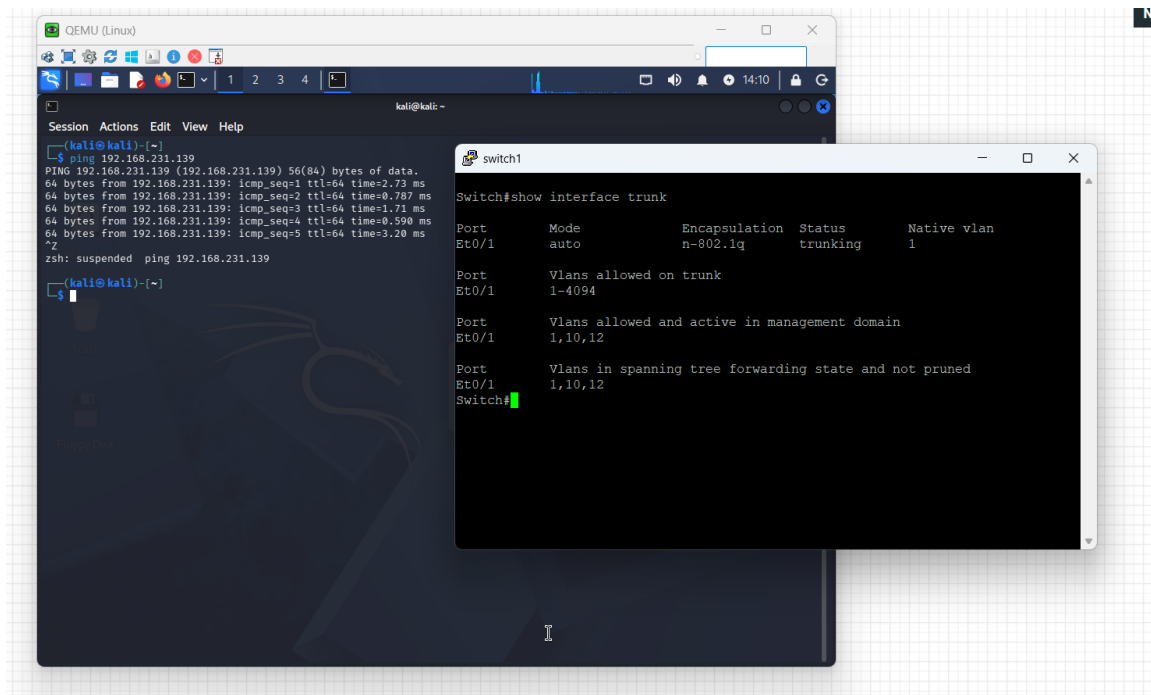
Now I will send DTP packet and then iwill check the ports status againe :



Lets cheick againe :



As we can see , our port now is trunk , whats that mean?? That's mean that now we are able to see all other vlans ,lets try ping metaspolaitable againe



Finally we can see all vlans , and the attack was done seccesfully